

KEYWORDS, IDENTIFIERS AND VARIABLES IN C

1. What is a Keyword?

A Keyword is a special word that already has a predefined meaning in C. These words are reserved by the compiler for specific purposes.

Definition: Keywords are reserved words in C that have predefined meanings and cannot be used as identifiers.

Keyword	Purpose
int	Integer variable
float	Decimal variable
char	Character variable
if	Decision making
else	Alternative decision
for	Loop
while	Loop
break	Exit loop
continue	Skip iteration
return	Return value
void	No value
static	Static variable/function
const	Constant value

Note : Traditionally, ANSI C contains 32 keywords.

2. What is an Identifier?

An Identifier is a name given by the programmer to identify variables, functions, arrays, and structures.

Real-World vs. C Programming Identifiers

Real Life: Gaurav (Student Name), Tommy (Dog Name), Delhi (City Name)

In C: age (Variable), calculate_salary (Function), marks_list (Array)

3. What is a Variable?

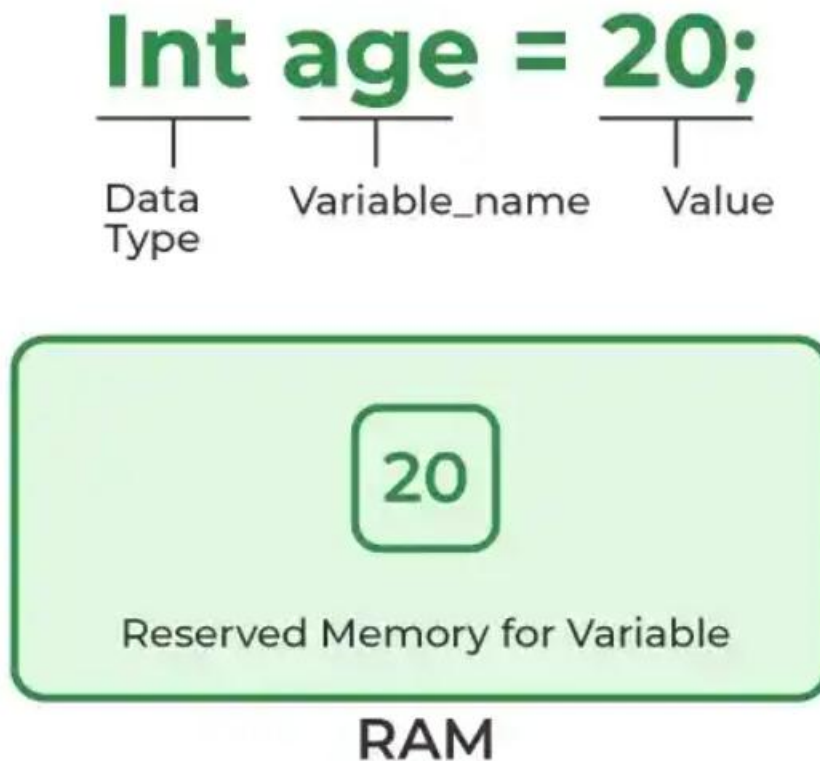
Variables are a subset of identifiers used to store data values.

Definition: A variable is the name of a memory location used to store data that can change during program execution.

Why is it Called a Variable?

Because its value can vary (change).

```
int age = 20;  
age = 21;  
age = 22;
```



Variable Declaration

Syntax:

`datatype variable_name;` **Note:** Datatypes are discussed in the next PDF (03)

Example:

```
int age;
```

Variable Initialization

Giving a value to a variable is called Initialization.

Example:

```
int age = 20;
```

Variable Naming Rules

These rules apply strictly to all identifiers (variables, functions, etc.).

The 5 Golden Rules of Naming

Rule 1: Start correctly * *Must* begin with a letter (a-z, A-Z) or an underscore (_). It **cannot** start with a number.

✓ salary, _name | ✗ 1age, 2salary

Rule 2: No Spaces * Spaces are completely illegal. Use camelCase or underscores instead.

✓ student_age, studentAge | ✗ student age

Rule 3: No Special Characters * Do not use symbols like @, #, %, \$.

✓ marks | ✗ marks%, age@

Rule 4: Keywords cannot be used* You cannot use keyword as a variable name

✗ int int; or float if;

Rule 5: Case Sensitivity * Uppercase and lowercase letters are treated as completely different entities.

age, Age, and AGE are **three separate variables**.

Types of Variables in C

1. Local Variable
2. Global Variable
3. Static Variable

Local Variable

A variable declared inside a function or block is called a Local Variable.

Global Variable

A variable declared outside the main() function is called a Global Variable.

Static Variable

A variable declared using the static keyword is called a Static Variable.

Note: Detailed behavior of Local, Global and Static variables will be studied later.